AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 9 · WORKBOOK

# Algorithms as Recipes

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

## One connected execution record.

Trace updates, distinguish replacement, repair an error and compare equal outputs.

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**Keep one habit:** Keep the instructions, initial state and every intermediate value.

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MAIN LESSON · SECTION 1

## Recall: reconnect earlier ideas

> Retrieve before opening notes.

### Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

**My remembered idea, example and correction**

**My remembered idea, example and correction**

▼ **Earlier wording, alternatives and picture transcripts**

**Features: Useful Clues**

**Sound Becomes a Signal**

**Sensors Are Messengers**

## W1 · Trace each current value

> Show the start and every arithmetic update.

### Your evidence record

- Start score at 3. Add 4 to score, double score, then subtract 5. Record the **initial state** and the **state** after each instruction. Underline the number used by the doubling **step** and explain where it came from.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 13

## W2 · Distinguish set from add

> Compare both procedures from the same initial value.

### Your evidence record

- Two procedures start score at 6. Procedure A sets score to 4, then adds 2. Procedure B adds 4 to score, then adds 2. **Trace** both and explain why the first instructions are different.

My reasoning and evidence

MAIN LESSON · SECTION 14

## W3 · Repair the earliest wrong transition

> Recompute later values from the corrected state.

### Your evidence record

- A procedure starts score at 6, adds 2, doubles score, then subtracts 3. A **trace** says 6, 8, 12, 9. Identify the first wrong transition, explain the likely mistake and write a complete corrected **trace**.

  Why is changing only the last value inadequate?

My reasoning and evidence

MAIN LESSON · SECTION 16

## W5 · Compare procedures with equal outputs

> Use intermediate states to check whether the required steps were followed.

### Your evidence record

- The required procedure starts score at 10, adds 2 and subtracts 2. A learner instead adds 4 and subtracts 4. Both report 10.

  Write both **trace**s and explain whether matching the final result proves that the learner followed the required procedure.

**My reasoning and evidence**

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MAIN LESSON · SECTION 4

## W4 · Trace two changing number boxes

> Keep one first attempt before hints or feedback.

### Read and explain independently

For a new display, X starts at 4 and Y starts at 1. Copy the current X into Y; add 3 to X; double Y; report both and stop. Record X and Y initially and after each instruction. Explain why Y does not become 7 when X changes.

Then swap the first two instructions and **trace** the new procedure from the same **initial state**.

**My independent explanation, trace and limits**

## Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

**Independent case: Trace two changing number boxes**

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MAIN LESSON · SECTION 9

# Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

## Your evidence record

Write the first executable sorting **algorithm** for one easy item before adding branches or AI.

The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

## Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

### Your evidence record

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

# Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

## Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next **step** _____

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

### Use this when helpful

Use W2: Distinguish replacement from addition for **Algorithm**s and changing **state**. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

## Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

### Use this when helpful

Use the worked case for **Algorithm**s and changing **state**. Proposed explanation: “Every instruction uses the original number”. Decide whether this explanation is supported.

Point to the **step** or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

## Reference · Original learning target

> Use this target to inspect an explanation.

## Use this when helpful

On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. **Trace** position and direction after every instruction. What if TURN were unspecified?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

# Extra practice · A complete case

> Try the question before checking the worked explanation.

## Use this when helpful

A robot starts on the left of three squares, facing right. It executes MOVE, MOVE, STOP, each MOVE being one square.

**Trace** the position after each command.

Show your **trace**, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ Extra practice · A changed case

REFERENCE · SECTION 7

## Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

### Use this when helpful

Instructions say pick up a pencil, write your name, then find paper. Find the first ordering problem and rewrite the three **step**s so another learner can follow them.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

**Week 9 Test and reflect**

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

## Extra practice · Design a revealing test

> Predict before tracing or testing.

## Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

# Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

## Use this when helpful

On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. **Trace** position and direction after every instruction. What if TURN were unspecified?

**My explanation after practice**

Record support used: none, hint or teacher explanation.

▼ **Reference · Independent-case response copy**

REFERENCE · SECTION 15

# Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

## Use this when helpful

For a new display, X starts at 4 and Y starts at 1. Copy the current X into Y; add 3 to X; double Y; report both and stop. Record X and Y initially and after each instruction. Explain why Y does not become 7 when X changes.

Then swap the first two instructions and **trace** the new procedure from the same **initial state**.

**My reasoning and evidence**

▼ **Extra practice · Use the worked picture**

REFERENCE · SECTION 18

# Extra practice · Use the worked picture

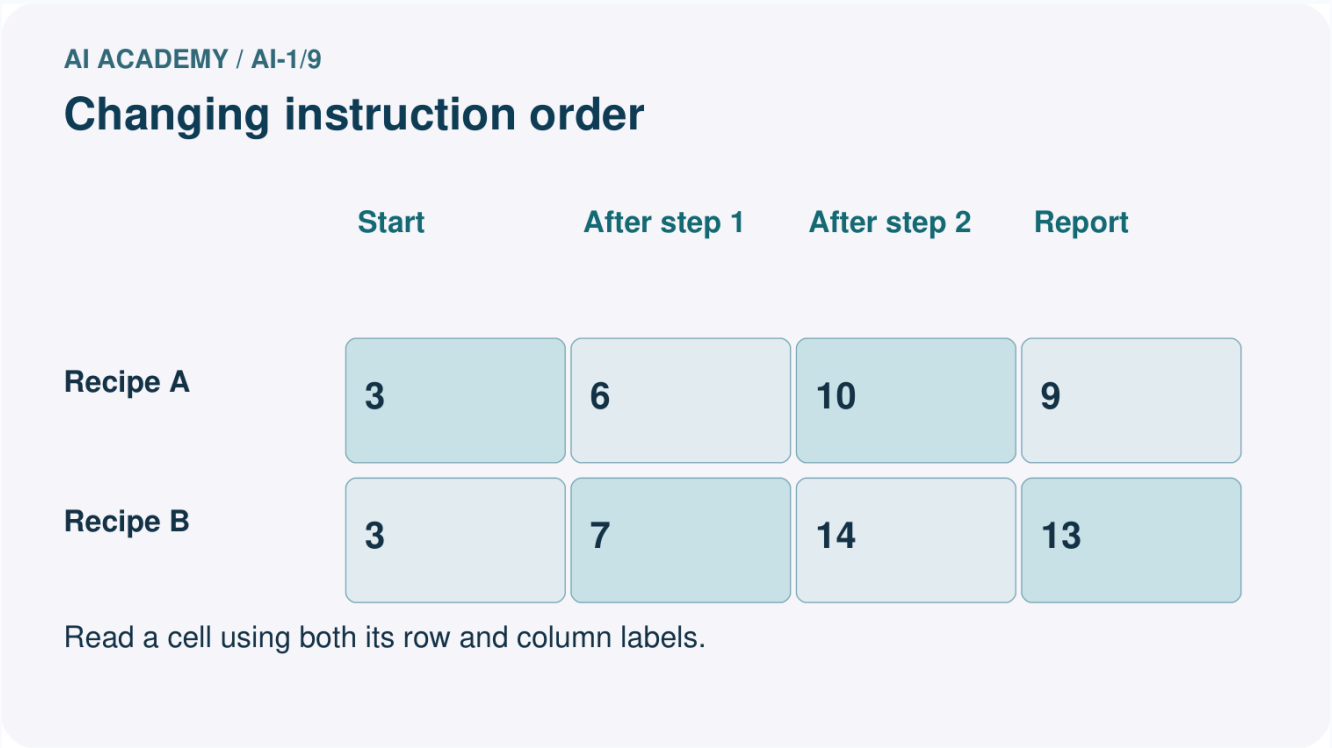
> Keep the explanation and the support you used.

## Use this when helpful

A fictional paper badge machine transforms an initial score using an ordered recipe.

Start score=3. Recipe A: double, add 4, subtract 1, report.

Recipe B swaps the first two operations but keeps the same starting score.




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/9

## Algorithms as Recipes

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

**1. Start**

Both runs begin at 3.

**2. Recipe A**

3 -> 6 after doubling -> 10 after adding 4 -> 9 after subtracting 1.

**3. Recipe B**

3 -> 7 after adding 4 -> 14 after doubling -> 13 after subtracting 1.

**4. Compare outputs**

Same operations, different order: outputs 9 and 13.

**CHANGE ONE THING**

Change the starting score to 0 for both recipes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Use the arrows to explain where the two zero-start runs first differ.

**My picture-based explanation and changed-case prediction**

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Write the sorter's **step**s in order, showing which values each **step** updates so a partner can reproduce the result.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

# Reference · Project portfolio record

> Extend your main project response when helpful.

## Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a **trace** of how one **input** becomes an **output**; explain the rule or learned component.

Write the sorter's **step**s in order, showing which values each **step** updates so a partner can reproduce the result.

| Evidence to keep | My record |
|---|---|
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

# Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

## First supported question

- A fictional paper badge machine transforms an initial score using an ordered recipe.
- Which labelled **step** matches this detail? Same operations, different order: **output**s 9 and 13.
- 1. Recipe A
- 2. Compare **output**s
- 3. Recipe B
- 4. Start
- Hint if needed: Read the **step**s in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare **output**s Same operations, different order: **output**s 9 and 13. Every instruction reads the current score, not the original one unless explicitly told to.

In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling. The final difference of 4 follows from that order, not from different starting data.

## Changed-case question

- Changed-case practice: Change the starting score to 0 for both recipes. Which conclusion is supported by the supplied facts?
- 1. A reports 3 and B reports 7.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding **step** in the pictorial **trace**. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: A reports 3 and B reports 7. A: 0→0→4→3. B: 0→4→8→7.

## Repair and save your work

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a **state**d task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered **sequence** of sound-signal samples with a **state**d time interval. Short practice: Does the all-zero retained record prove the full signal was silent?

Use the picture.. Return to this week and explain what you changed.

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

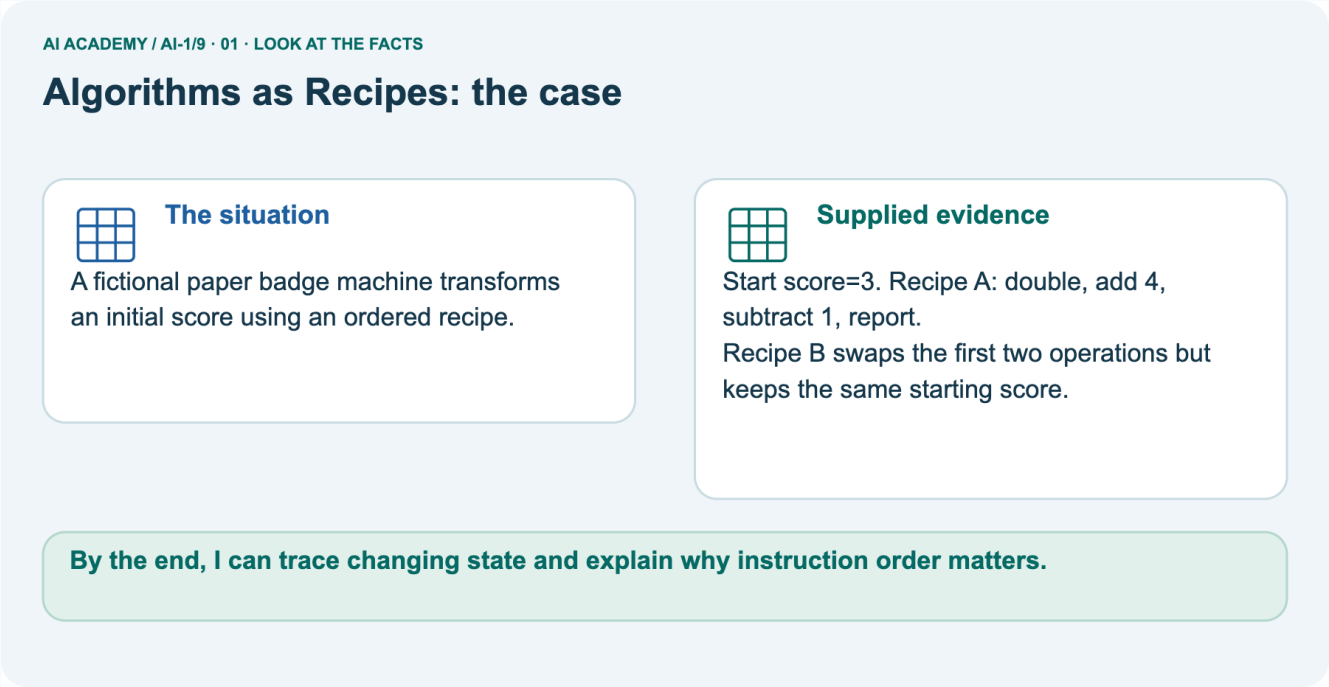
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

## Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

# 01. Algorithms as Recipes: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

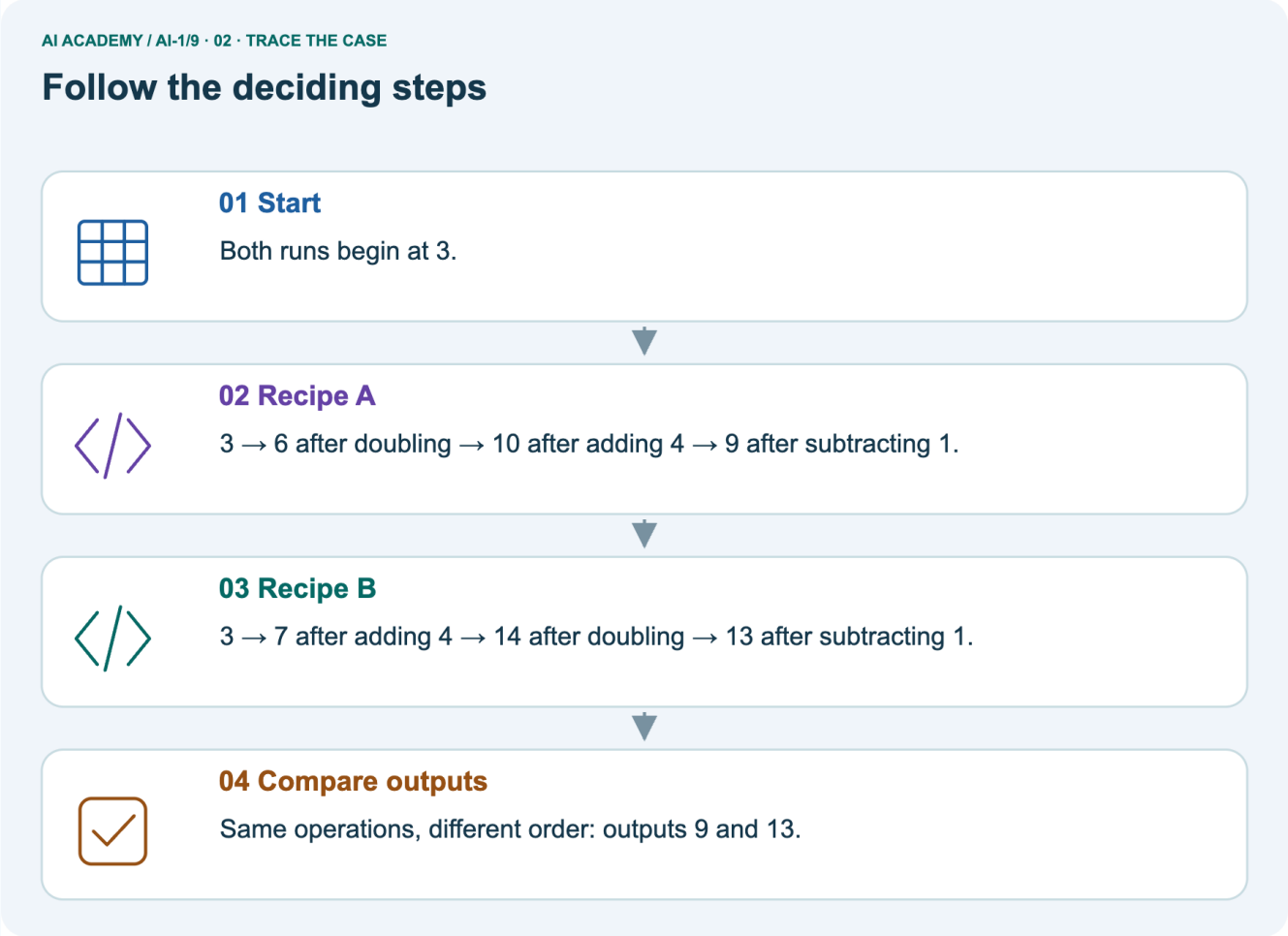
A fictional paper badge machine transforms an initial score using an ordered recipe. Supplied facts: Start score=3. Recipe A: double, add 4, subtract 1, report.

Recipe B swaps the first two operations but keeps the same starting score.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



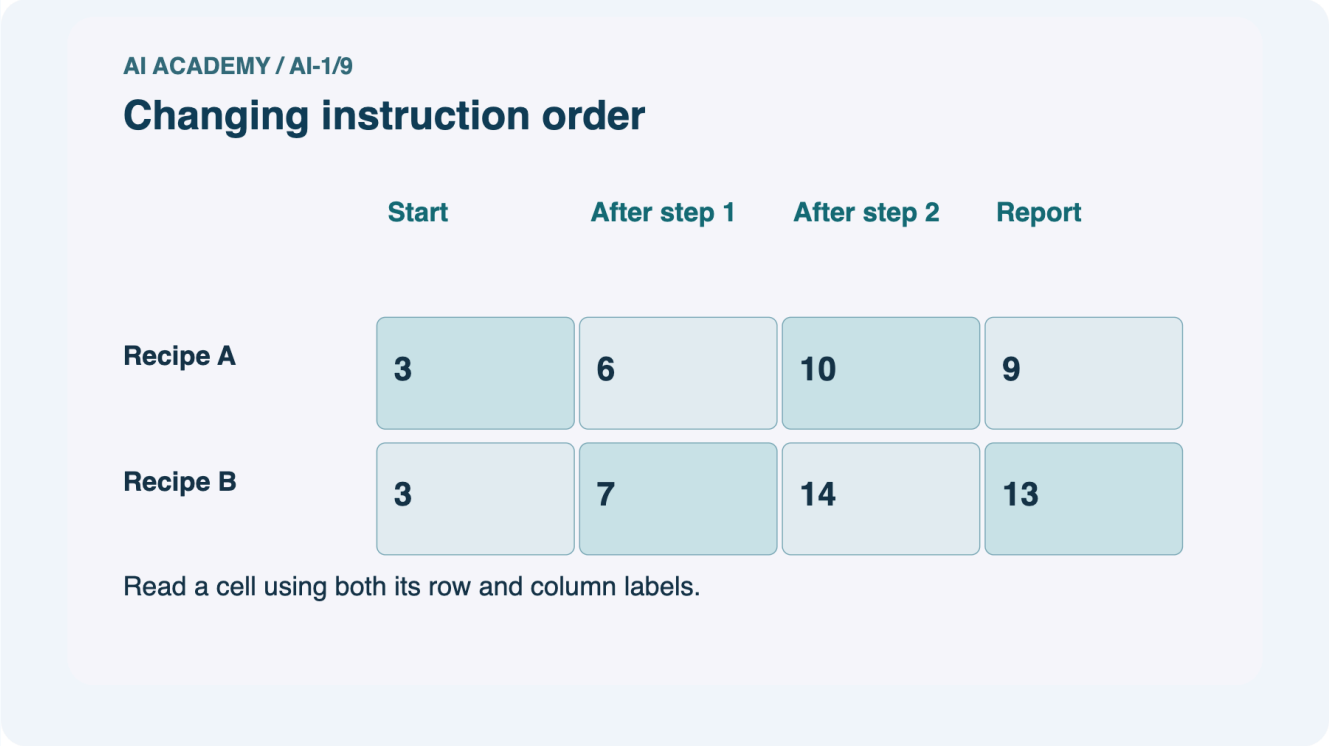

Follow the labelled picture with the matching explanation. Open the vector diagram

Start: Both runs begin at 3. → Recipe A: 3 → 6 after doubling → 10 after adding 4 → 9 after subtracting 1. → Recipe B: 3 → 7 after adding 4 → 14 after doubling → 13 after subtracting 1. → Compare **output**s: Same operations, different order: **output**s 9 and 13.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Changing instruction order



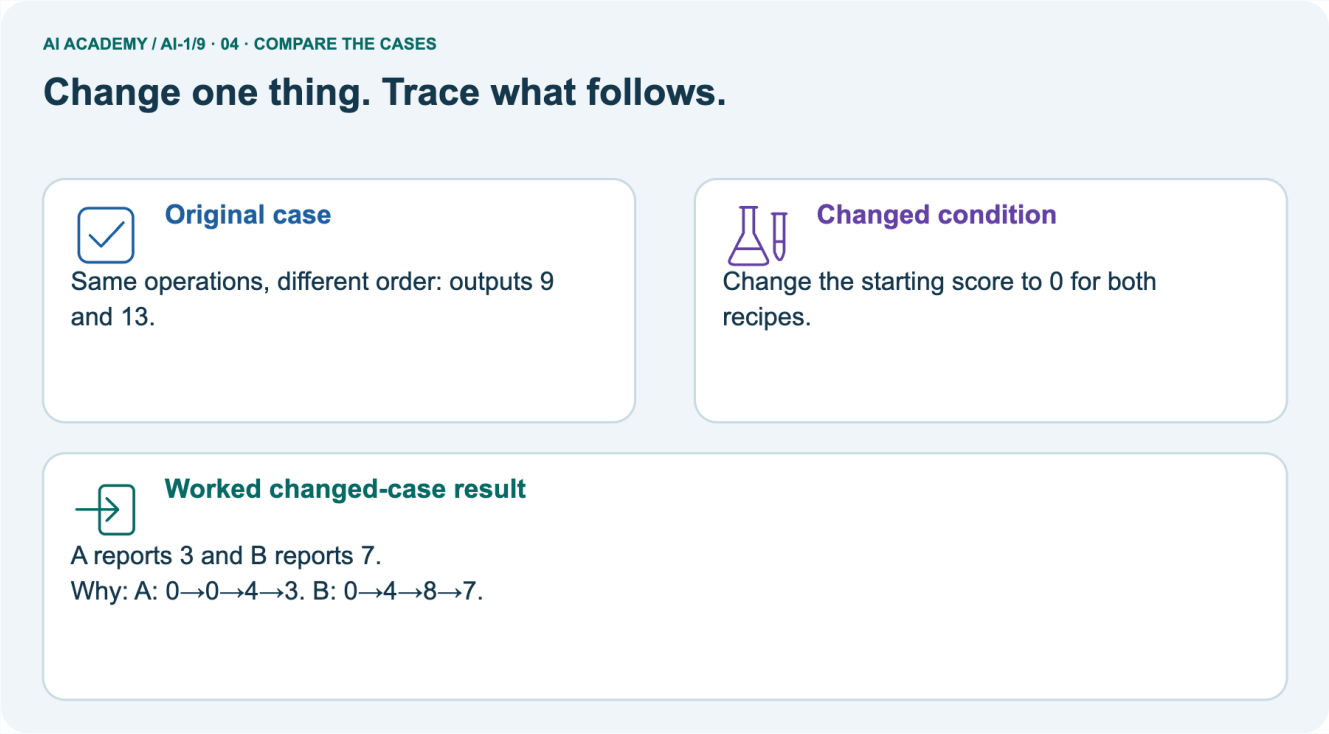

Follow the labelled picture with the matching explanation. Open the vector diagram

Changing instruction order. Recipe A: Start = 3, After **step** 1 = 6, After **step** 2 = 10, Report = 9. Recipe B: Start = 3, After **step** 1 = 7, After **step** 2 = 14, Report = 13.

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.



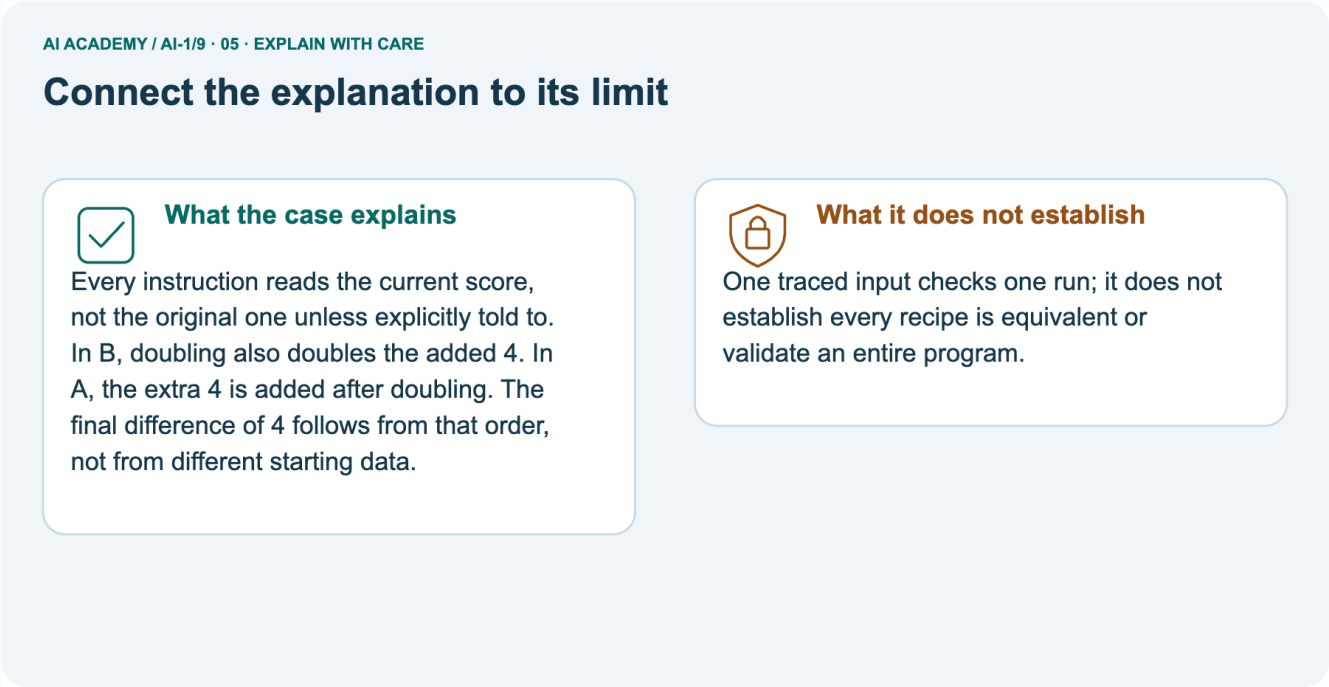

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Same operations, different order: **output**s 9 and 13. Changed condition: Change the starting score to 0 for both recipes. Expected result: A reports 3 and B reports 7. Reason: A: 0→0→4→3. B: 0→4→8→7.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

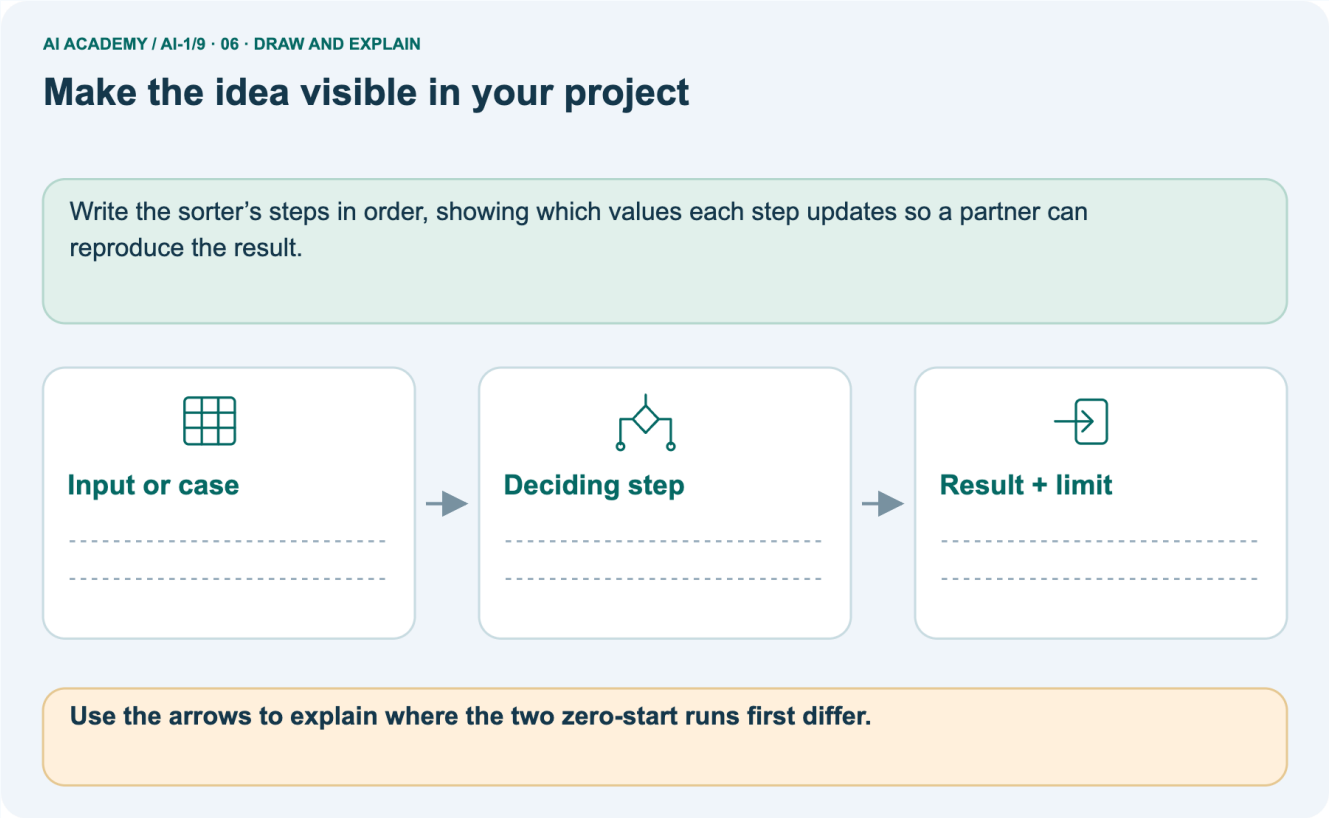
Every instruction reads the current score, not the original one unless explicitly told to. In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling.

The final difference of 4 follows from that order, not from different starting data. Limit: One **trace**d **input** checks one run; it does not establish every recipe is equivalent or validate an entire program.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Write the sorter's **step**s in order, showing which values each **step** updates so a partner can reproduce the result. Draw your **input**, deciding **step**, result and limit.

Supported question: Use the arrows to explain where the two zero-start runs first differ.

Draw your project **trace** and justify the deciding **step** using **evidence**.

### My picture explanation, deciding evidence and one limit

## ▼ Earlier wording, alternatives and picture transcripts

### 01. Algorithms as Recipes: the case

### 02. Follow the deciding steps

### 03. Changing instruction order

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.